Fueling the Fare: Gas Prices and Subway Ridership in NYC

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Abstract

In an era of volatile energy markets and climate-related transit goals, policymakers seek greater knowledge of how these two objectives connect. Using a standard time-series model, this study attempts to isolate and evaluate the impact of gasoline prices on subway ridership in New York City. Using a standard linear regression to establish correlation and a log-log Ordinary Least Squares (OLS) model to estimate predictive impact, we measure an elasticity of 0.3 (p ¡ 0.001), meaning a 1% increase in gas prices predicts a 0.3% increase in ridership. Our model addresses potential biases through a deterministic time trend and robust covariance estimation. However, diagnostic testing reveals substantial positive autocorrelation (Durbin-Watson of 0.824), indicating high persistence of transit habits. Policy implications include supply-side interventions such as improved transit quality and removing barriers to transit-oriented development potentially yielding larger ridership results.

1 Introduction

Transit choices are among the most consequential, and the most path-dependent, ones that a society will make. The United States is uniquely car-dependent relative to peer countries, in part due to low density allowing for less population compression. The key exception is New York City. With a density of approximately $11,313.8$ people/km^2, it is among the most crowded areas in America, providing the necessary agglomeration to make large-scale public transportation spatially efficient. To these ends, it possesses one of the most developed subway systems globally, with 399 km of track, 472 stations and approximately 1.3 billion annual riders in 2025. Based on 2021 data, 32 percent of New Yorkers use the subway as their primary means of transportation to work, with 54 percent of New Yorkers using it at least weekly.

Current literature largely focuses on nationwide or statewide impacts of gasoline prices on transportation choices. We believe that localized urban data in cities with robust transit networks is highly effective at estimating elasticities due to agglomeration effects driving more robust public transportation infrastructure. Using data from the Metropolitan Transit Authority and Energy Information Administration on subway daily ridership (which is compressed into weekly averages) and gas prices within New York City, we seek to further the body of research on gas prices in areas with diverse transit choices.

2 Data

The dataset consists of information from late June 2021 until the start of April 2026. The start point of late June 2021 was chosen due to New York State's COVID-19 emergency declaration officially ending on June 24, 2021. 2021 also was a time when gas prices consistently trended upwards as COVID-19 era mobility constraints ended. To account for that potential confounding variable, we use a 4-week moving average for ridership and correct heteroskedacity using a Heteroskedasticity-Consistent 3 (HC3) method.

3 Methodology

We decided to use a log-log model to establish cleaner predictive modelling. To see whether or not two things are substitutable, it is necessary to model elasticities. The simplest method, without losing accuracy given only two variables, is a log-log model.

Given the volatility of commodity markets, we expect:

$$\mathrm{Corr}(\Delta p_t, \Delta p_{t-k}) \approx 0$$

Yet consumer habit persistence implies:

$$\rho \to 1$$

in the residual process, consistent with our Durbin-Watson statistic of 0.824.

In English, a large degree of heteroskedasticity (variance in errors between different levels of the independent variable) is to be expected. Consumer expectations play a key role in transit decisions, and in volatile markets like commodities, prices are less autocorrelated week-to-week than behavior.

One limitation of this study is the lack of formal modeling of how commuters form expectations about gas prices. We can only state that, given the chart in Figure 1, there is likely a lag between an initial price spike and changes in transportation methods.

4 Results

Table 1: Model Diagnostic Statistics

Dep. Variable:	ln_ridership	R-squared:	0.591
Model:	OLS	Adj. R-squared:	0.587
Method:	Least Squares	F-statistic:	131.0
No. Observations:	243	Prob (F-statistic):	3.47e-39
Df Residuals:	240	AIC:	-431.7
Df Model:	2	BIC:	-421.2
Covariance Type:	HC3	Log-Likelihood:	218.83

Table 2: OLS Regression Results

	Coefficient	Std. Error	z	P> $\lvert z \rvert$	0.025	0.975
const	14.3342	0.098	145.677	0.000	14.141	14.527
ln_gas_price	0.2954	0.067	4.412	0.000	0.164	0.427
trend	0.0019	0.000	14.297	0.000	0.002	0.002

Table 3: Residual Diagnostics

Omnibus:	21.643	Durbin-Watson:	0.824
Prob(Omnibus):	0.000	Jarque-Bera:	24.709
Skew:	-0.758	Prob(JB):	4.31e-06
Kurtosis:	3.375	Cond. No.:	2.55e+03

The log-log OLS model yields an elasticity of 0.2954, meaning a 1% increase in gas prices predicts approximately a 0.30% increase in subway ridership, significant at the 0.1% level. The model explains 59.1% of variance in log-ridership ($R^2 = 0.591$), with the time trend accounting for secular post-COVID ridership recovery.

The residual plot reveals approximate zero-mean errors with no severe heteroskedastic pattern, validating our HC3 correction. However, several outliers suggest idiosyncratic shocks not captured by the model, consistent with the elevated Jarque-Bera statistic ($p = 4.31 \times 10^{-6}$). The Durbin-Watson statistic of 0.824 indicates substantial positive autocorrelation, suggesting that ridership in

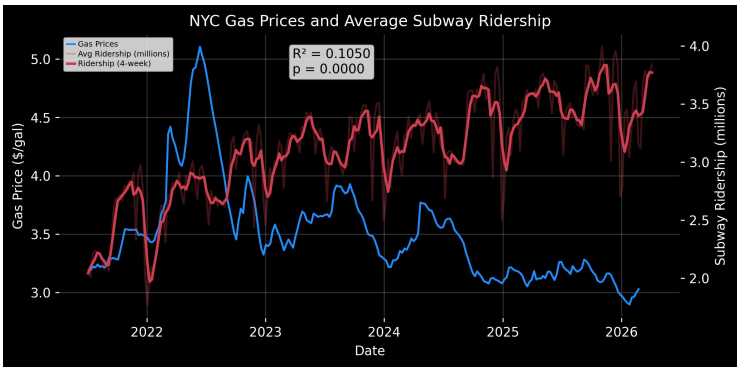

Figure 1: Raw correlation between gas prices and ridership.

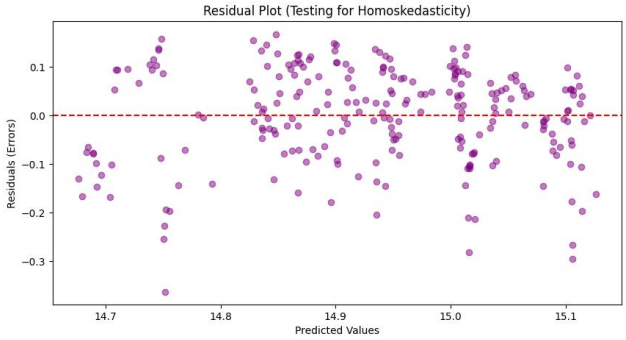

Figure 2: Residual plot.

a given week is strongly predicted by the prior week — consistent with habitual commuting behavior.

5 Policy Implications

Our findings suggest that gas prices exert a statistically significant but modest influence on subway ridership. An elasticity of 0.30 implies that large price swings such as the 2022 spike exceeding $5 per gallon are required to produce meaningful ridership shifts. This limits the reliability of gas prices as a policy lever for increasing transit usage.

The high degree of autocorrelation in ridership further implies behavioral inertia: commuters are slow to change transportation habits even in response to price signals. This suggests that supply-side interventions such as improving service frequency, reliability, and coverage are likely more effective at increasing ridership than relying on gas price volatility.

Altogether, these findings suggest gas-tax revenue or other policy interventions to price gas above prior market equilibria may be effective if directed toward improvements for existing stations. Additionally, lowering barriers to transit-oriented development may serve to provide both for greater ridership and to provide a robust tax base to continue funding other municipal services.

6 Conclusion

This study estimates a gas price elasticity of subway ridership of 0.30 in New York City from June 2021 to April 2026, using a log-log OLS model with HC3-robust standard errors and a deterministic time trend. While the effect is statistically robust, its modest magnitude and the high persistence of ridership habits suggest that commuter behavior is more inertial than price-sensitive.

Several limitations warrant consideration. The Durbin-Watson statistic of 0.824 heavily indicates unresolved autocorrelation, which a formal AR(1) or Newey-West correction could address. Additionally, our model does not account for seasonality, fare changes, service disruptions, or broader macroeconomic conditions. Future work could extend this analysis to other transit modes or cities, or incorporate an expectations model for how commuters anticipate price changes.

7 Appendix

Source Code and Files (GitHub)